In [1]:
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# Name: Hemant Marotkar ,UID: - 23015002
"""
Experiment: Evaluate classification models using Confusion Matrix, ROC Curve and AU
Cyber Security Context: Classify network traffic as Normal (0) or Attack (1)
Required libraries:
pip install pandas scikit-learn matplotlib
"""
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report,
    roc_curve,
    auc,
    roc_auc_score
)
# ------------------------------------------------------------
# 1. Load cyber security dataset
# ------------------------------------------------------------
df = pd.read_csv("cyber_classification_eval_dataset.csv")
print("First five rows of dataset:")
print(df.head())
# ------------------------------------------------------------
# 2. Separate input features and target label
# ------------------------------------------------------------
# X contains traffic features such as packet count, bytes, failed logins, etc.
# y contains class label:
# 0 = Normal traffic
# 1 = Attack traffic
X = df.drop("attack", axis=1)
y = df["attack"]
# ------------------------------------------------------------
# 3. Split dataset into training and testing sets
# ------------------------------------------------------------
# stratify=y keeps equal proportion of Normal and Attack records
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42,
    stratify=y
)
# ------------------------------------------------------------
# 4. Define classification models
# ------------------------------------------------------------
models = {
    "Logistic Regression": Pipeline([
        ("scaler", StandardScaler()),
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        ("classifier", LogisticRegression())
    ]),
    "Random Forest": RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )
}
# ------------------------------------------------------
# 5. Train, predict and evaluate each model
# ------------------------------------------------------
plt.figure(figsize=(8, 6))
for model_name, model in models.items():
    print("\n" + "=" * 60)
    print("Model:", model_name)
    # Train model
    model.fit(X_train, y_train)
    # Predict class labels
    y_pred = model.predict(X_test)
    # Predict probability score for Attack class
    y_score = model.predict_proba(X_test)[:, 1]
    # Confusion Matrix
    cm = confusion_matrix(y_test, y_pred)
    print("\nConfusion Matrix:")
    print(cm)
    # Classification Report
    print("\nClassification Report:")
    print(classification_report(y_test, y_pred, target_names=["Normal", "Attack"]))
    # AUC Score
    auc_score = roc_auc_score(y_test, y_score)
    print("AUC Score:", round(auc_score, 3))
    # Save confusion matrix image
    disp = ConfusionMatrixDisplay(
        confusion_matrix=cm,
        display_labels=["Normal", "Attack"]
    )
    disp.plot()
    plt.title(f"Confusion Matrix - {model_name}")
    plt.savefig(f"confusion_matrix_{model_name.replace(' ', '_').lower()}.png")
    plt.close()

    # ROC Curve
    fpr, tpr, thresholds = roc_curve(y_test, y_score)
    roc_auc = auc(fpr, tpr)

    plt.plot(fpr, tpr, label=f"{model_name} AUC = {roc_auc:.3f}")
# ------------------------------------------------------
# 6. Plot and save combined ROC curve
# ------------------------------------------------------
plt.plot([0, 1], [0, 1], linestyle="--", label="Random Guess")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve for Cyber Attack Classification")
plt.legend()
plt.grid(True)
plt.savefig("roc_curve_models.png")
plt.show()
```

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First five rows of dataset:
   duration_ms  packet_count  failed_login_attempts  src_bytes  dst_bytes  \
0          120            15                      2        300        400
1           95            10                      1        250        350
2         3000           250                     20       9000      12000
3         2800           230                     18       8500      11000
4          130            14                      1        280        370

   urgent_packets  protocol_tcp  protocol_udp  attack
0               0             1             0       0
1               0             1             0       0
2              12             0             1       1
3              10             0             1       1
4               0             1             0       0

============================================================
Model: Logistic Regression

Confusion Matrix:
[[5 0]
 [0 4]]

Classification Report:
              precision    recall  f1-score   support

      Normal       1.00      1.00      1.00         5
      Attack       1.00      1.00      1.00         4

    accuracy                           1.00         9
   macro avg       1.00      1.00      1.00         9
weighted avg       1.00      1.00      1.00         9

AUC Score: 1.0

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Model: Random Forest

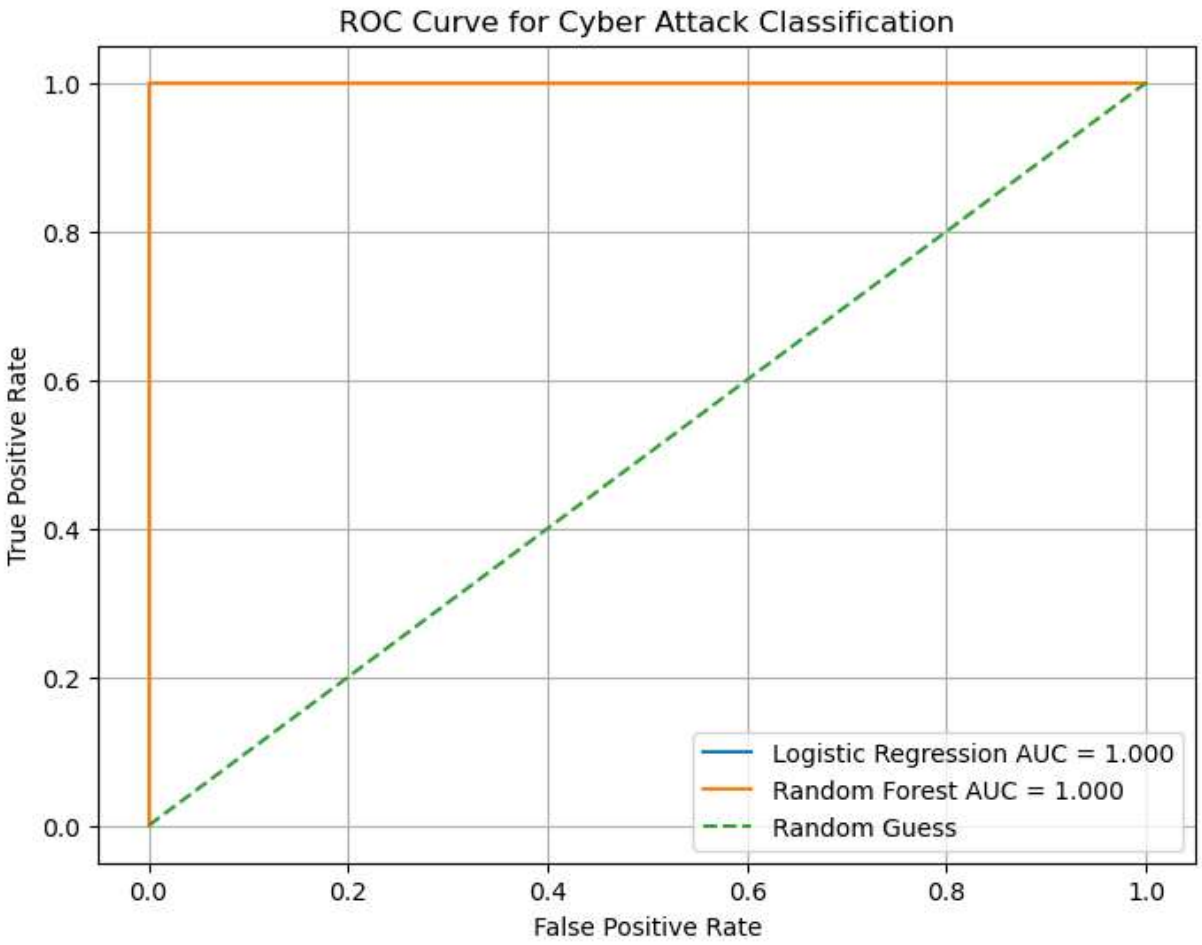
Confusion Matrix:
[[5 0]
 [0 4]]

Classification Report:
              precision    recall  f1-score   support

      Normal       1.00      1.00      1.00         5
      Attack       1.00      1.00      1.00         4

    accuracy                           1.00         9
   macro avg       1.00      1.00      1.00         9
weighted avg       1.00      1.00      1.00         9

AUC Score: 1.0
```


ROC Curve for Cyber Attack Classification
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression AUC = 1.000
Random Forest AUC = 1.000
Random Guess